Theoretical Astroparticle Physics

(PHYS708)

Self introduction

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RESEARCH INTEREST/FIELD

Astronomy, Astrophysics and Astroparticle Physics (theory and experiment):
- Cosmic rays, high-energy gamma rays, astrophysical neutrinos
- Member of the LOFAR, SKA, ALTO and KM3NeT collaborations

09/2002 - 03/2009: **RESEARCH SCIENTIST**
Astrophysical Sciences Division, Bhabha Atomic Research Centre, Mumbai, India.

04/2009 - 03/2012: **Ph.D (Astrophysics and Astroparticle Physics)**
Department of Astrophysics, Radboud University Nijmegen, Netherlands.

04/2012 - 07/2015: **POSTDOCTORAL RESEARCHER**
Department of Astrophysics, Radboud University Nijmegen, Netherlands.

08/2015 - 03/2019: **POSTDOCTORAL RESEARCHER**
Department of Physics and Electrical Engineering, Linnaeus University, Växjö, Sweden.

08/2019 - Present: **ASSISTANT PROFESSOR**
Physics Department, Khalifa University, UAE.

Overview of the course

Assessment Methodology			
		Tentative Dates	**Weight**
Coursework:	Assignments	See page 3-4	10%
	Quiz 1	See page 3-4	10%
	Quiz 2	See page 3-4	10%
Projects (if applicable)	One project with approval	See page 3-4	20%
Semester Examination(s)	Midterm test	See page-4	20%
Final Examination	Final test	TBA (registrar office)	30%

Topics to be covered (tentative)

Teaching Plan (Tentative):

Week 1 Aug 25 - Aug 29	Lecture 1: Introduction to high-energy Astrophysical objects. Lecture 2: Messengers in high-Energy Astrophysics.
Week 2 Sep 1 - Sep 5	Lecture 3: Interaction of high-energy particles I (Hadronic processes)
Week 3 Sep 8 - Sep 12	Lecture 3: Interaction of high-energy particles I (Hadronic processes) contd. Lecture 4: Interaction of high-energy particles II (Leptonic processes) *Assignment 1 (due Sep 12)*
Week 4 Sep 15 - Sep 19	Lecture 5: Particle acceleration: First and Second-order Fermi acceleration
Week 5 Sep 22 - Sep 26	Lecture 6: Particle acceleration at astrophysical shocks
Week 6 Sep 29 - Oct 3	Lecture 7: High-energy emissions (Gamma rays, neutrinos and secondary electrons/positrons). *Assignment 2 (due Oct 3)*
Week 7 Oct 6 - Oct 10	Lecture 7: High-energy emissions (Gamma rays, neutrinos and secondary electrons/positrons) cont.

Topics to be covered (tentative)

Week 8 Oct 13 - Oct 17	*Mid-term exam week (tentative)*
Week 9 Oct 20 - Oct 24	Lecture 8: Propagation of high-energy particles I (Cosmic-ray protons and nuclei) *Assignment 3 (due Oct 24)*
Week 10 Oct 27 - Oct 31	Lecture 9: Propagation of high-energy particles II: Electrons. *Assignment 4 (due Oct 31)*
Week 11 Nov 3 - Nov 7	Lecture 10: Some open problems in cosmic-ray physics.
Week 12 Nov 10 - Nov 14	Lecture 11: Experiments in Astroparticle Physics: Space-based experiments. *Assignment 5 (due Nov 14)*
Week 13 Nov 17 - Nov 21	Lecture 12: Experiments in Astroparticle Physics: Ground-based experiments
Week 14 Nov 24 - Nov 28	Project week (tentative).
Week 15 Dec 1 - Dec 5	Project week (tentative). Oral presentation of project results.
Dec 6 - Dec 12	*Final Exam period*

Main books/literatures to be followed

High Energy Astrophysics (3rd edition)
Malcolm S. Longair, Cambridge University Press (2011), ISBN: 9780521756181.

Cosmic gamma rays
Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288.

Very High Energy Cosmic Gamma radiation
F A. Aharonian, World Scientific Publishing (2004), ISBN 9789812561732.

Cosmic rays and Particle Physics (2nd edition)
Thomas K. Gaisser, Ralph Engel and Elisa Resconi, Cambridge University Press (2016)
ISBN: 9781139192194.

Radiative Processes in Astrophysics,
George B. Rybicki and Alan P. Lightman,
Print ISBN:9780471827597   Online ISBN:9783527618170.

Several journal papers which will be mentioned during the lectures.

Astronomy Today
Chaisson & McMillan
(This book is used only for the some basic introduction)